

DIRECTIONS *for questions 1 to 5:* Answer the questions on the basis of the information given below.

In a castle, there are seven dungeons, D1 through D7, which can be opened or closed by operating four levers, L1 through L4. Each time a lever is operated once, any dungeon which is operated (i.e., can be opened or closed) by that lever will open, if it was initially closed, OR will close, if it was initially open.

L1 operates the dungeons D1, D3, D5 and D6. L2 operates the dungeons D2, D4 and D7. L3 operates the dungeons D3, D4, D5 and D6. L4 operates the dungeons D1, D2, D4 and D5.

All the dungeons are initially closed and there are seven prisoners, Adam, Bob, Charles, Donald, Evan, Frank and Gary, in the seven dungeons, not necessarily in the same order, such that there is exactly one prisoner in each dungeon. Any prisoner can escape from the dungeon which he is in, if and only if that dungeon is open.

Further, the following information is known about the way the levers operate:

- i. When all the dungeons are closed, if L2 and another lever are each operated exactly once, Evan and Gary are among the ones who cannot escape.
- ii. When all the dungeons are closed, if L1 and another lever are each operated exactly once, only Adam, Bob, Evan and Frank can escape.
- iii. When all the dungeons are closed, if L3 and two other levers are each operated exactly once, Bob, Donald and Frank are among the ones who can escape.

Q1. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

Who is in D4?

- ☐ a) **Charles**
- ☐ b) **Evan**
- ☐ c) **Frank**
- ☐ d) **Adam**

Q2. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.
What BEST can be said about the dungeon in which Donald is present?

- ☐ a) **D5**
- ☐ b) **D6**
- ☐ c) **D7**
- ☐ d) **Either D1 or D7**

Q3. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

When all the dungeons are closed, if each of the four levers is operated exactly once, who among the following definitely cannot escape?

- ☐ a) **Bob**
- ☐ b) **Frank**
- ☐ c) **Charles**
- ☐ d) **More than one of the above**

Q4. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

If Bob is one of the persons who can escape when L3 is operated exactly once when all the dungeons are closed, in which dungeon is Frank?

- ☐ a) **D3**
- ☐ b) **Either D3 or D5**
- ☐ c) **D5**
- ☐ d) **Either D3 or D6**

Q5. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

When all the dungeons are closed, if each of exactly three levers is operated exactly once, exactly two persons can escape. Which of the following options BEST represents one or more of the two persons who can escape?

- ☐ a) **Charles**
- ☐ b) **Bob**
- ☐ c) **Donald**
- ☐ d) **Both Charles and Donald**

DIRECTIONS for questions 6 to 10: Answer the questions on the basis of the information given below.

Hari is the operations manager in a factory that manufactures toys. During a certain month, he experimented with five different processes – Process I through Process V – for manufacturing toys in his factory. The table below provides, for each of the five processes, partial information about the manufacturing cost incurred per unit manufactured using that process, the number of units manufactured during that month using that process and the total manufacturing cost incurred using that process. Each row in the table represents the information for one process, not necessarily in any order.

Manufacturing Cost per unit (in ₹/unit)	Number of units	Total Manufacturing Cost (in ₹)
39		
	16500	6,60,000
21	19000	
		7,65,000
25		3,75,000

The following information is known about the units manufactured during the month using the five processes:

- Considering all the units manufactured during the month, the average cost of manufacturing one unit was ₹31.80
- The number of units manufactured using Process III was 2000 more than the number of units manufactured using Process IV, the cost per unit for which was the second lowest.
- The cost per unit was the third highest for Process V and the total cost for Process III was not greater than ₹7,50,000.
- The cost per unit for Process II was less than that for Process V, while the cost per unit for Process III was at least ₹35.

Q6. DIRECTIONS for question 6: Type in your answer in the input box provided below the question.

How many units were manufactured during the month using Process III?

Q7. DIRECTIONS *for question 7:* Select the correct alternative from the given choices.
The total manufacturing cost during the month was the highest for

- ☐ a) **Process I.**
- ☒ b) **Process V.** ✓ **Your answer is correct**
- ☐ c) **Process II.**
- ☐ d) **Process III.**

Q8. DIRECTIONS *for question 8:* Type in your answer in the input box provided below the question.
What is the total number of units manufactured during the given month using all the five processes?

Q9. DIRECTIONS *for questions 9 and 10:* Select the correct alternative from the given choices.

If the total number of units that Hari manufactured in the next month is the same as that in the given month and he manufactured all the units using the process for which the cost per unit is the lowest, what will be the total cost of manufacturing for the next month?

- ☒ a) ₹18,90,000 ✓ **Your answer is correct**
- ☐ b) ₹19,11,000
- ☐ c) ₹17,57,500
- ☐ d) ₹17,10,000

Q10. DIRECTIONS *for questions 9 and 10:* Select the correct alternative from the given choices.

What is the manufacturing cost per unit for Process II?

- ☒ a) ₹21 ✓ Your answer is correct
- ☐ b) ₹25
- ☐ c) ₹20
- ☐ d) ₹24

DIRECTIONS *for questions 11 to 15:* Answer the questions on the basis of the information given below.

Mohan collected coins and notes of four different currencies – Senzu, Lingot, Urmi and Tristan. It is known that, in each of these four currencies, there are only two denominations – one denomination in the form of a coin and the other in the form of a note. In each currency, one coin represents one unit of that currency. For example, the value of one coin in Senzu currency is equal to 1 Senzu. Further, in each currency, the value of each note is an even multiple of the value of each coin in that currency. Mohan had at least three notes of each currency.

The following information is also known:

- i. The value of three Senzus is same as the value of two Lingots and the value of two Urmis is same as the value of three Tristans.
- ii. The total value of the coins and notes that he had of each currency was the same for all four currencies.
- iii. He had eight Lingot notes and seven Tristan notes. The number of Urmi notes that he had was not more than five.
- iv. He had a total of 66 coins with him. The number of Lingot coins he had was same as the number of Urmi coins he had and the number of Tristan coins he had was ten more than the number of Senzu coins he had.
- v. The total amounts of Senzus and Urmis he had with him are 72 Senzus and 52 Urmis respectively.
- vi. He had a total of 28 notes. The value of a Tristan note is eight times that of a Tristan coin.

Q11. DIRECTIONS *for questions 11 and 12:* Type in your answer in the input box provided below the question.

How many Senzu notes did Mohan have with him?

Q12. DIRECTIONS *for questions 11 and 12:* Type in your answer in the input box provided below the question.

The value of how many Senzu coins is equal to that of 13 Urmi notes?

Q13. DIRECTIONS *for questions 13 and 14:* Select the correct alternative from the given choices.

The value of all the Lingot coins that Mohan had is equal to that of how many Senzu notes?

☐ a) 4

☐ b) 5

☐ c) 6

☐ d) 7

Q14. DIRECTIONS *for questions 13 and 14:* Select the correct alternative from the given choices.

If Mohan exchanged 13 Tristan coins that he had for Lingot notes with one of his friends, what would be the sum of the number of Urmi notes and the number of Lingot notes with him, after the exchange?

☐ a) 16

☐ b) 20

☐ c) 13

☐ d) 15

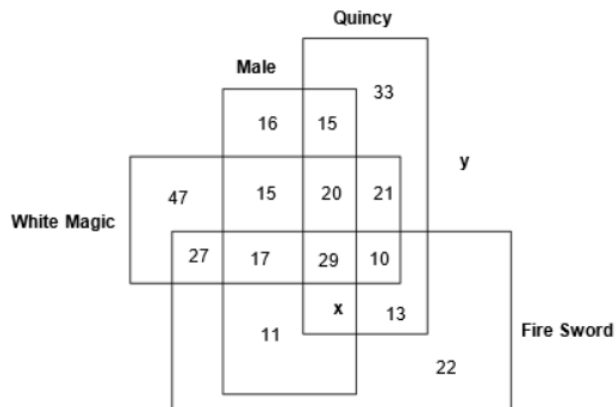
Q15. DIRECTIONS *for question 15:* Type in your answer in the input box provided below the question.

If all the currency that he had are converted to Lingots, what would be the total amount (in Lingots) that he had with him?

DIRECTIONS for questions 16 to 20: Answer the questions on the basis of the information given below.

On the magical island of Gumpville, there are a total of 360 islanders, such that for every two male islanders, there are three female islanders. Every islander belongs to exactly one of the two tribes, Shinigami and Quincy. Each islander practises exactly one of the two types of magic prevalent on the island, Dark Arts and White Magic. Further, each islander owns exactly one mystic weapon between an Ice Shield and a Fire Sword.

The following Venn Diagram illustrates the number of islanders belonging to different categories:



Q16. DIRECTIONS for question 16: Type in your answer in the input box provided below the question.

How many islanders practice Dark Arts and own a Fire Sword?

Q17. DIRECTIONS *for question 17:* Select the correct alternative from the given choices.

How many islanders belonging to the Shinigami tribe either own an Ice Shield or practice White Magic but not both?

- ☐ a) 103
- ☐ b) 69
- ☐ c) 96
- ☐ d) 107

Q18. DIRECTIONS *for question 18:* Type in your answer in the input box provided below the question.

What is the difference between the number of females belonging to the Shinigami tribe and owning Ice Shields and the number of males belonging to the Quincy tribe and practicing White Magic?

Q19. DIRECTIONS *for questions 19 and 20:* Select the correct alternative from the given choices.

What is the total number of females who neither own an Ice Shield nor belong to the Quincy tribe?

☐ a) 49

☐ b) 22

☐ c) 72

☐ d) 57

Q20. DIRECTIONS *for questions 19 and 20:* Select the correct alternative from the given choices.

What is the difference between the number of males who own Fire Swords and the number of females who own Ice Shields?

Note:The correct answer for question 20 in the **DILR** section of this **AIMCAT** is 66, which was not present among the answer choices during the exam. Therefore, this question has been ignored for evaluation and marks have been awarded to all students on a pro-rata basis. Now it has been corrected

☐ a) 78

☐ b) 66

☐ c) 82

☐ d) 80